6TH GRADE MATH · UNIT 5: INEQUALITIES

# Theme Park Rules

*Teacher Guide & Answer Keys*

**Teacher**

2–3 class periods

## Project Overview

Students plan a theme park day. Ride rules become inequalities, budgets and time limits become one-step inequalities, and students design rules for a new ride.

Graphing on a number line shows the solution set and the meaning of open and closed endpoints.

## Standards Addressed

- 6.EE.B.5 Understand solving an inequality as finding values that make it true
- 6.EE.B.8 Write inequalities x > c or x < c and graph them
- 6.EE.B.7 Solve one-step equations and related inequalities
- Honors: combine two inequalities to describe a range

## Pacing (2–3 class periods)

| Day | Plan |
|---|---|
| Day 1 | Launch with real ride signs. Part 1 rules as inequalities and graphs. |
| Day 2 | Part 2 budget and time inequalities. |
| Day 3 | Part 3 design a ride; build the day plan. |

## Materials

- Student handout and Ride Rule Cards
- Measuring tape (optional)
- Calculators
- Poster paper

## How the Three Versions Differ

| Level | What changes |
|---|---|
| CP2 | Single rules, open vs. closed circles, and one-step inequalities with starters. |
| CP1 | Several riders and rides, ranges between two rules, and budget inequalities with rounding. |
| Honors | Compound conditions, impossible combinations, multi-step constraints, and engineering rules that meet several conditions. |

## Watch For (Common Misconceptions)

- Using an open circle for ≥ or ≤.
- Rounding up when the context says "at most".
- Reversing the inequality symbol when writing from words.

## Extension Ideas

- Measure classmates (with permission) and make a class ride chart.
- Research real ride height rules.
- Graph two rules on one number line in different colors.

# Answer Key – CP2

## Part 1: Ride Rules

| # | Task | Answer |
|---|---|---|
| 1 | Write an inequality for the Thunder Coaster rule. | $h \geq 48$ |
| 2 | Graph the Thunder Coaster rule on a number line. Use a closed circle or an open circle? | Closed circle at 48, shade to the right. |
| 3 | Write and graph the Kiddie Cars rule. | $h < 42$; open circle at 42, shade left. |
| 4 | Maya is 47 inches tall. Which rides can she go on? | Sky Swings (≥ 44) and Log Flume (≥ 46). Not Thunder Coaster, Drop Tower, or Kiddie Cars. |

## Part 2: Budget Check

| # | Task | Answer |
|---|---|---|
| 5 | Game tickets cost $3 each. You can spend at most $24. How many tickets t can you buy? | $3t \leq 24 \rightarrow t \leq 8$ tickets |
| 6 | You have $40. After lunch you have to keep at least $15. Lunch costs x dollars. Write an inequality. | $40 - x \geq 15 \rightarrow x \leq 25$ |
| 7 | A ride takes 5 minutes. You have 45 minutes. How many times r can you ride? | $5r \leq 45 \rightarrow r \leq 9$ rides |
| 8 | Is 10 a solution of $3t \leq 24$? Explain. | No, because 3(10) = 30 and 30 is not ≤ 24. |

## Part 3: Plan Your Day

| # | Task | Answer |
|---|---|---|
| 9 | Pick three rides. List the height rule for each and say who in a group of heights 43, 46, 49, and 53 inches can ride. | Answers vary. E.g., Thunder Coaster: 49 and 53. Log Flume: 46, 49, 53. Sky Swings: 46, 49, 53. |
| 10 | Write an inequality for how much you can spend on snacks if snacks cost $4 each and your snack budget is $16. | $4s \leq 16 \rightarrow s \leq 4$ snacks |
| 11 | Make a schedule for your day. Show that your total time on rides is at most 3 hours (180 minutes). | Answers vary; check that the total minutes ≤ 180. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | When do you use a closed circle, and when do you use an open circle? | Closed for ≤ or ≥ (the number is included); open for < or > (not included). |

| # | Prompt | Look for |
|---|---|---|
| R2 | Why does an inequality have many solutions? | Any number in the shaded region makes the inequality true, not just one value. |

# Answer Key – CP1

## Part 1: Who Can Ride?

| # | Task | Answer |
|---|---|---|
| 1 | Write each ride rule as an inequality (h for height, w for weight, a for age). | Thunder $h \geq 48$; Kiddie $h < 42$; Sky Swings $h \geq 44$ and $w \leq 220$; Bumper Boats $a \geq 8$; Log Flume $h \geq 46$; Drop Tower $h \geq 52$ |
| 2 | Make a table showing which rides each friend can go on. | Ava: Sky Swings. Ben: Thunder, Sky Swings, Log Flume. Cruz: Kiddie Cars. Dev: Thunder, Sky Swings, Log Flume. No one can ride the Drop Tower; Bumper Boats depends on age, not height. |
| 3 | Graph the heights that can ride the Sky Swings but NOT the Log Flume. | $44 \leq h < 46$: closed circle at 44, open circle at 46, shade between. |
| 4 | Ben grows $\frac{1}{2}$ inch per month. How many more months until he can ride the Drop Tower? | $51 + 0.5m \geq 52 \rightarrow 0.5m \geq 1 \rightarrow m \geq 2$ months |

## Part 2: Money & Time

| # | Task | Answer |
|---|---|---|
| 5 | Lunch costs $11.50. Game tickets cost $2.50. Write and solve an inequality for the number of tickets t you can buy. | $11.50 + 2.50t \leq 40 \rightarrow 2.50t \leq 28.50 \rightarrow t \leq 11.4 \rightarrow$ 11 tickets (solving after subtracting lunch is one step) |
| 6 | You have 2 hours left at the park. Each coaster ride takes 40 minutes including the wait. Write and solve an inequality for the number of rides r. | $40r \leq 120 \rightarrow r \leq 3$ rides |
| 7 | The park says a group needs at least 12 people for a discount. Your group has 7. Write and solve an inequality for how many more people p you need. | $7 + p \geq 12 \rightarrow p \geq 5$ |
| 8 | Explain why t = 11.4 is not a sensible answer for game tickets. | You cannot buy part of a ticket; the greatest whole number that works is 11. |

## Part 3: Design a Ride

| # | Task | Answer |
|---|---|---|
| 9 | Write a height rule for your ride with a minimum AND a maximum (two inequalities). Graph it. | Answers vary; e.g., $h \geq 40$ and $h \leq 60$ shown with closed circles and shading between. |
| 10 | Using the four friends above, who can ride your ride? | Answers vary based on the rule. |
| 11 | Each ride car holds at most 450 pounds. If each rider weighs 90 pounds, how many riders can share a car? Write and solve an inequality. | $90r \leq 450 \rightarrow r \leq 5$ riders |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | How do you decide whether to round a solution up or down? | Read the situation: "at most" limits round down; "at least" requirements round up. |
| R2 | How is the solution of an inequality different from the solution of an equation? | An equation usually has one solution; an inequality has a set (range) of solutions. |

# Answer Key – Honors

## Part 1: Ride Logic

| # | Task | Answer |
|---|---|---|
| 1 | Write the heights that can ride the Log Flume but not the Drop Tower. Graph the set. | $46 \le h < 52$ |
| 2 | Write the heights that can ride neither the Kiddie Cars nor the Sky Swings. | $42 \le h < 44$ |
| 3 | Is there a height that can ride every ride? Explain using inequalities. | No: Kiddie Cars needs $h < 42$ but Drop Tower needs $h \ge 52$; no number is both. |
| 4 | A rider is 4 feet 1 inch tall and weighs 105 pounds. List every ride they can ride. | 49 in: Thunder Coaster, Sky Swings ($w \le 220$), Log Flume. |

## Part 2: Constraints

| # | Task | Answer |
|---|---|---|
| 5 | Entry is \$28. Snacks cost \$3.75. You want to keep at least \$5 for the bus. Write and solve an inequality for snacks $s$. | $28 + 3.75s + 5 \le 50 \rightarrow 3.75s \le 17 \rightarrow s \le 4.53 \rightarrow$ 4 snacks |
| 6 | A group discount gives \$4 off each ticket for groups of at least 15. Your class of 23 needs at least how many more friends to form two discount groups? | 2 groups need at least 30: $23 + f \ge 30 \rightarrow f \ge 7$ |
| 7 | Wait time grows by 6 minutes every half hour after 10:00 a.m., starting at 10 minutes. You want to wait at most 40 minutes. Up to how many half hours after 10:00 can you get in line? | $10 + 6x \le 40 \rightarrow x \le 5 \rightarrow$ by 12:30 p.m. |
| 8 | Explain why the boundary point of a "≤" solution is included but a "<" boundary is not, using one of the problems above. | For snacks, $3.75s = 17$ would be exactly the money allowed, so the boundary is allowed; with "<" the exact amount would break the rule. |

## Part 3: Engineer a Ride

| # | Task | Answer |
|---|---|---|
| 9 | A ride car can hold at most 600 lb. The car and harness weigh 150 lb. Riders average 95 lb. Write and solve an inequality for the number of riders. | $150 + 95r \le 600 \rightarrow r \le 4.7 \rightarrow$ 4 riders |
| 10 | Write a height rule (minimum and maximum) so that at least three of the four friends (45, 51, 41, 48 in) can ride but the Drop Tower rule is still stricter. | Answers vary; e.g., $44 \le h \le 60$ lets Ava, Ben, Dev ride; Drop Tower ($\ge 52$) is stricter. |

| # | Task | Answer |
|---|---|---|
| 11 | Without a Fast Pass, 3 rides cost $5 each and each ride has a 35-minute wait. A Fast Pass for 3 rides skips the waits. Write an inequality for a Fast Pass price p that costs more than paying for the 3 rides but less than $20. Explain the trade-off. | $15 < p < 20$. Example: a $18 pass costs $3 more but saves $3 \times 35 = 105$ minutes of waiting. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | How can two simple inequalities describe a range? Give an example. | Use a minimum and a maximum together, such as $46 \le h < 52$. |
| R2 | Why is it impossible to satisfy some pairs of inequalities at the same time? | If their solution sets do not overlap on the number line, no number satisfies both. |